Chapter 3: Vector Calculus

The following subjects will be discussed

- Definition of Differentials in Different Coordinates.
- Integrals: Line, surface and volume integrals.
- The "del" Operator.
- Gradient, Divergence, Curl and Laplacian operators.

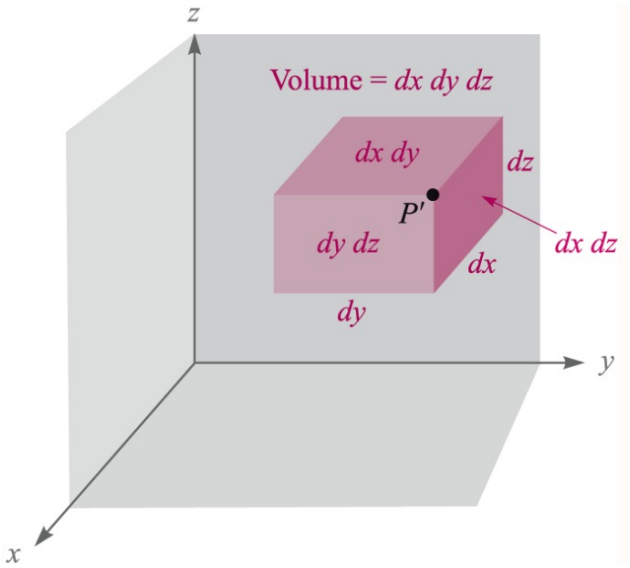
Differentials in Rectangular Coordinates

The differential length, area and volume in rectangular coordinate system are given by

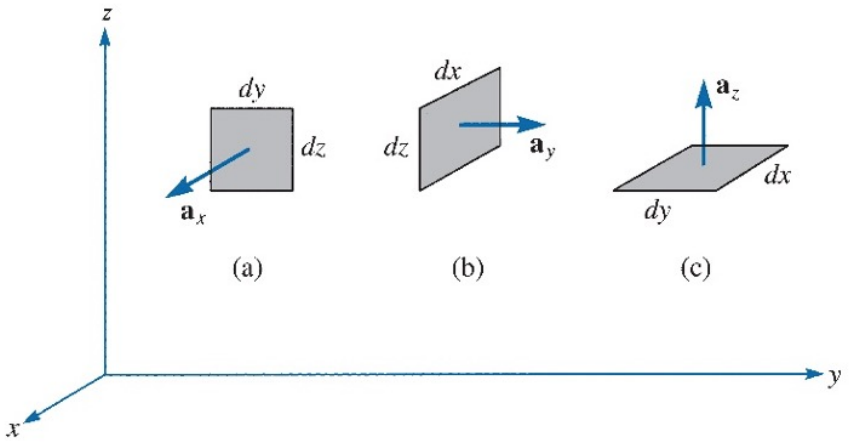
$$\overrightarrow{dl} = dx\,\vec{a}_x + dy\,\vec{a}_y + dz\,\vec{a}_z$$ Differential length

$$\overrightarrow{dS} = dy\,dz\,\vec{a}_x + dx\,dz\,\vec{a}_y + dx\,dy\,\vec{a}_z$$ Differential area

$$dv = dx\,dy\,dz$$ Differential volume



The differential area vector is defined normal to the area.



Differentials in Cylindrical Coordinates

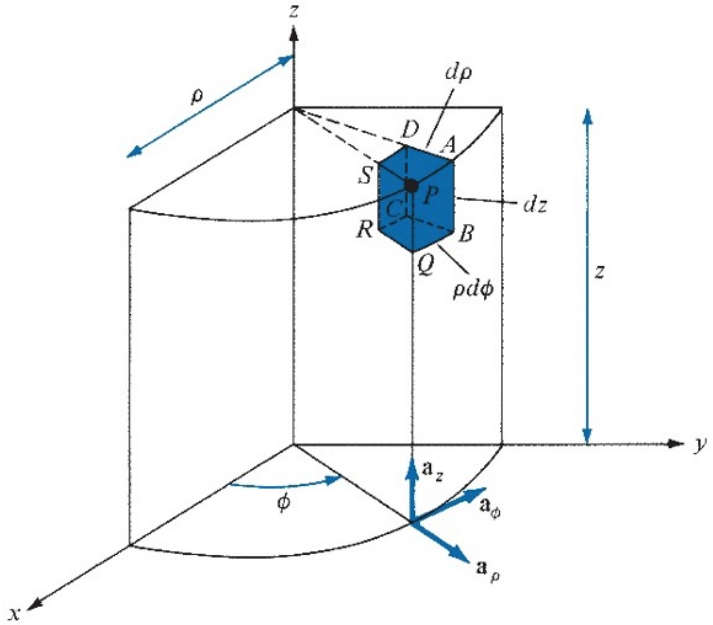
The differential length, area and volume in cylindrical coordinate system are given by

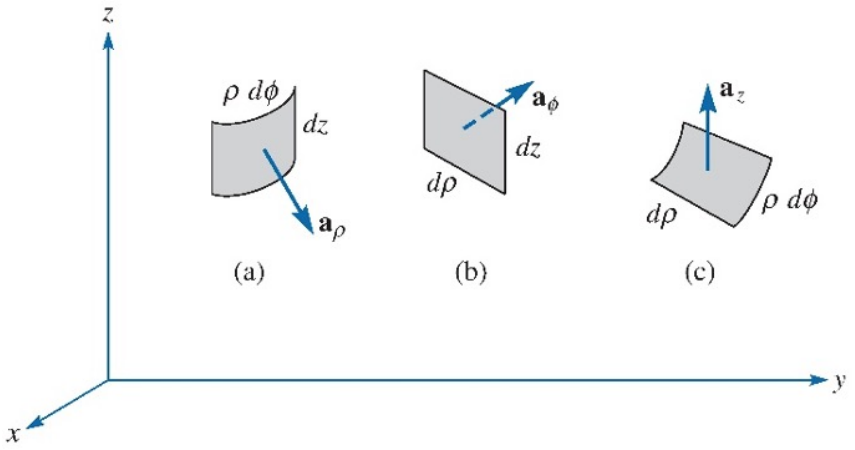
$$\overrightarrow{dl} = d\rho\,\vec{a}_\rho + \rho\,d\varphi\,\vec{a}_\varphi + dz\,\vec{a}_z$$ Differential length

$$\overrightarrow{dS} = \rho\,d\varphi\,dz\,\vec{a}_\rho + d\rho\,dz\,\vec{a}_\varphi + \rho\,d\rho\,d\varphi\,\vec{a}_z$$ Differential area

$$dv = \rho\,d\rho\,d\varphi\,dz$$ Differential volume

The differential area vector is defined normal to the area.





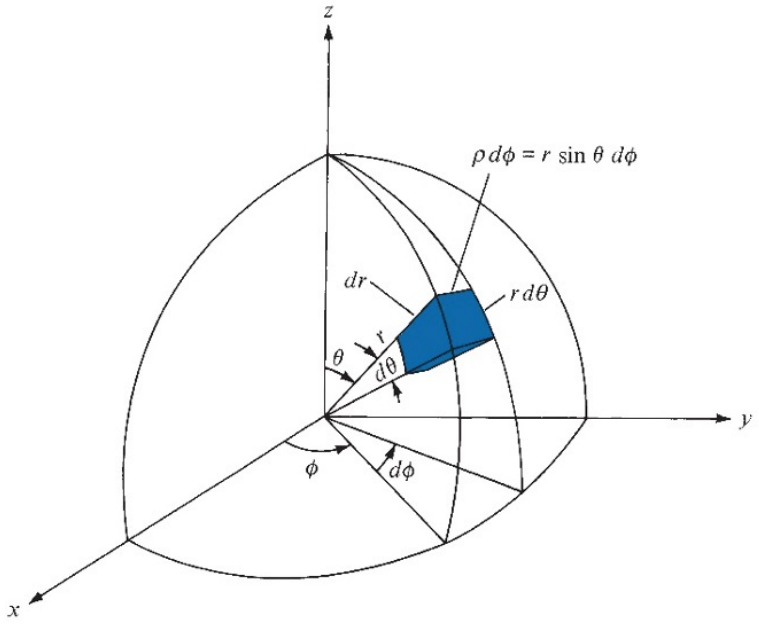
Differentials in Spherical Coordinates

The differential length, area and volume in spherical coordinate system are given by

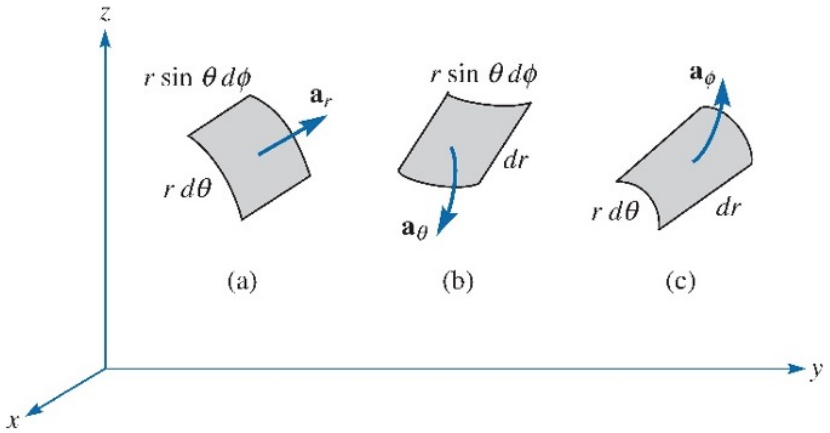
$$\overrightarrow{dl} = dr\,\vec{a}_r + r\,d\theta\,\vec{a}_\theta + r\sin\theta\,d\varphi\,\vec{a}_\varphi$$ Differential length

$$\overrightarrow{dS} = r^2\sin\theta\,d\theta\,d\varphi\,\vec{a}_r + r\sin\theta\,dr\,d\theta\,\vec{a}_\theta + r\,dr\,d\theta\,\vec{a}_\varphi$$ Differential area

$$dv = r^2\sin\theta\,dr\,d\theta\,d\varphi$$ Differential volume



The differential area vector is defined normal to the area.

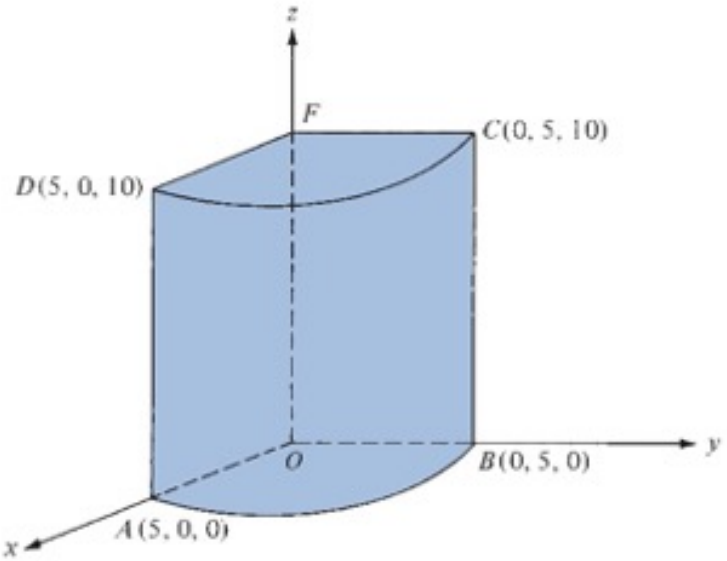


Example

Use cylindrical coordinates system. Dimensions in meters. Find:

The length of line AB.

$$\overrightarrow{dl} = d\rho\,\vec{a}_\rho + \rho\,d\varphi\,\vec{a}_\varphi + dz\,\vec{a}_z$$



$$\text{Length of AB} = \int_L dl = \int_0^{\pi/2} \rho\,d\varphi = \rho\int_0^{\pi/2} d\varphi = (5)\left(\frac{\pi}{2}\right) = 2.5\pi \ \text{(m)}$$

The area of surface ABCD.

$$\overrightarrow{dS} = \rho\,d\varphi\,dz\,\vec{a}_\rho + d\rho\,dz\,\vec{a}_\varphi + \rho\,d\rho\,d\varphi\,\vec{a}_z$$

$$\text{Area ABCD} = \iint_S dS = \iint \rho\,d\varphi\,dz = \rho\int_0^{\pi/2} d\varphi \int_0^{10} dz = (5)\left(\frac{\pi}{2}\right)(10) = 25\pi \ (\text{m}^2)$$

The volume of structure. $dv = \rho\,d\rho\,d\varphi\,dz$

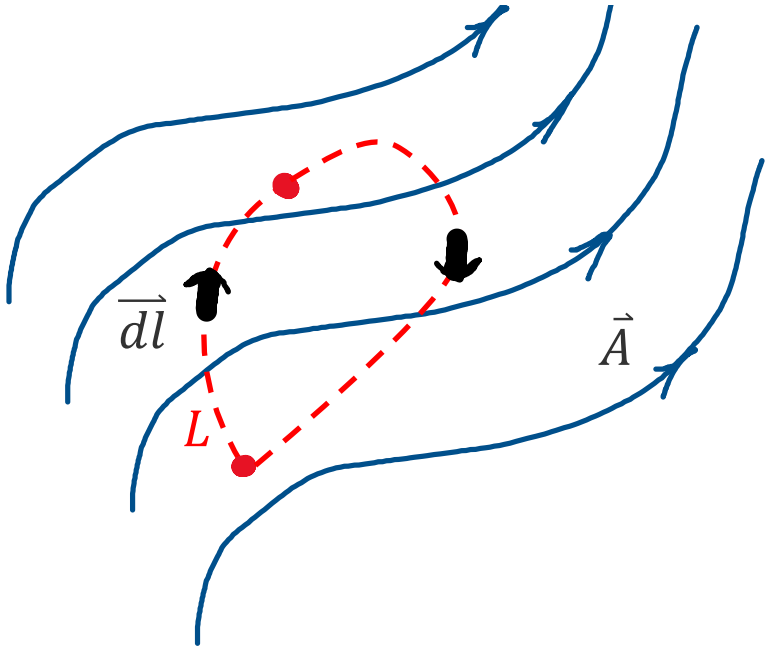
$$\text{Volume} = \int_V dv = \iiint \rho\,d\rho\,d\varphi\,dz = \int_0^5 \rho\,d\rho \int_0^{\pi/2} d\varphi \int_0^{10} dz = \left(\frac{25}{2}\right)\left(\frac{\pi}{2}\right)(10) = 62.5\pi \ (\text{m}^3)$$

Vector Integration: Line Integral

The line integral measures how much of the vector magnitude is in a certain direction.

$$\text{Line Integral} \equiv \int_L \vec{A} \cdot \overrightarrow{dl}$$

$$\text{Closed Line Integral} \equiv \oint_L \vec{A} \cdot \overrightarrow{dl}$$



Examples of common line integrals in physics include:

Potential difference: $V_{AB} = -\int_A^B \vec{E} \cdot \overrightarrow{dl}$

Work: $U_{AB} = \int_A^B \vec{F} \cdot \overrightarrow{dl}$

Ampere's Law: $I = \oint_L \vec{H} \cdot \overrightarrow{dl}$

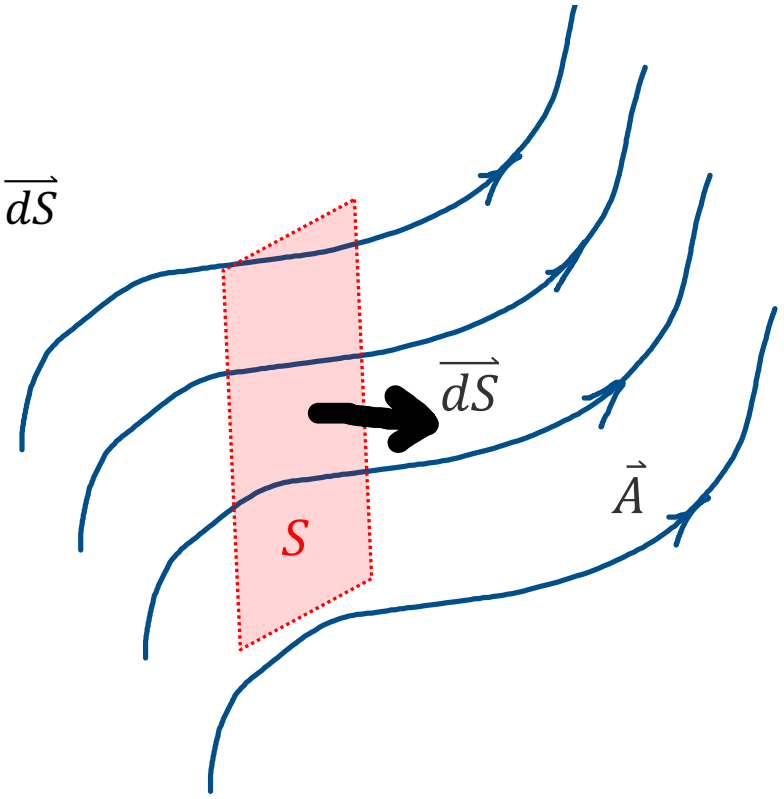
Note: The unit of the line integral result is: Units of $\vec{A}$ × meters

Vector Integration: Surface Integral

The surface integral measures vector flux through a given surface.

$$\text{Surface Integral (flux)} \equiv \iint_S \vec{A} \cdot \overrightarrow{dS}$$

$$\text{Closed Surface Integral (net flux)} \equiv \oiint_S \vec{A} \cdot \overrightarrow{dS}$$



Examples of common surface integrals in physics include:

Total current: $I = \iint_S \vec{J} \cdot \overrightarrow{dS}$

Magnetic flux: $\phi = \iint_S \vec{B} \cdot \overrightarrow{dS}$

Note: The unit of the surface integral result is: Units of $\vec{A} \times \text{meters}^2$

Volume Integral

The volume integral is performed on scalar density quantities.

$$\text{Volume Integral} \equiv \iiint_V (density)\, dv$$

Examples of common volume integrals in physics include:

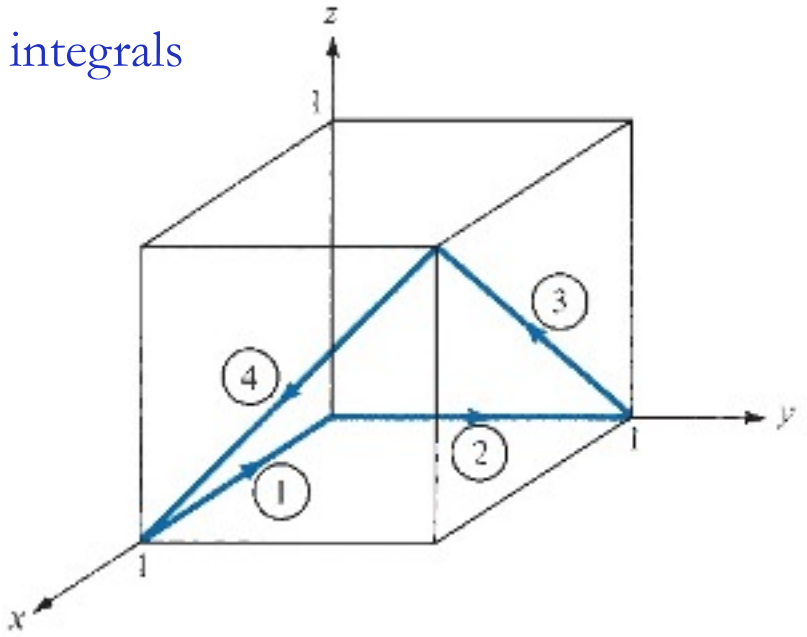
Total charge: $Q = \iiint_V \rho_v \, dv$

Total mass: $M = \iiint_V \rho_v \, dv$

Note: The unit of the volume integral result is: Units of density × meters^3

Examples

(1) Given $\vec{F} = x^2\,\vec{a}_x - xz\,\vec{a}_y - y^2\,\vec{a}_z$, find the line integrals along the indicated paths.



For L_1 $\quad \overrightarrow{dl} = dx\,\vec{a}_x + dy\,\vec{a}_y + dz\,\vec{a}_z$

$$\int_{L1} \vec{F} \cdot \overrightarrow{dl} = \int_1^0 x^2 dx = -\frac{1}{3}$$

For L_4 $\quad \overrightarrow{dl} = dx\,\vec{a}_x + dy\,\vec{a}_y + dz\,\vec{a}_z$

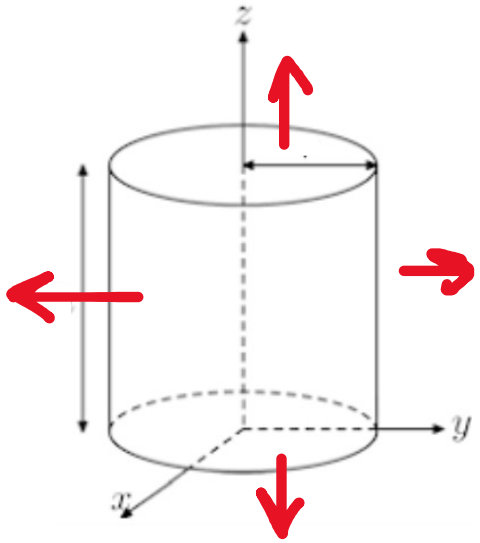
$$\vec{F} \cdot \overrightarrow{dl} = -xz\,dy - y^2\,dz$$

$$\int_{L4} \vec{F} \cdot \overrightarrow{dl} = -\int_1^0 xz\,dy - \int_1^0 y^2\,dz = -x\int_1^0 y\,dy - \int_1^0 z^2\,dz = (1)\left(\frac{1}{2}\right) + \left(\frac{1}{3}\right) = \frac{5}{6}$$

Examples

(2) Given $\vec{G} = 10e^{-2z}\left(\rho\,\vec{a}_\rho + \vec{a}_z\right)$ (C/m^2), find the total flux from a cylindrical surface of radius 1 m, height 1 m, with axis along positive z axis from z=0.

Surface Integral (flux) $\equiv \iint_S \vec{A}\cdot\overrightarrow{dS}$



For top surface $\overrightarrow{dS} = \rho\,d\rho\,d\varphi\,\vec{a}_z$

$$\vec{G}\cdot\overrightarrow{dS} = 10e^{-2z}\rho\,d\rho\,d\varphi$$

$$Flux = \iint_{top}\vec{G}\cdot\overrightarrow{dS} = 10e^{-2(1)}\int_0^1\rho d\rho\int_0^{2\pi}d\varphi = 10\pi e^{-2} \quad \text{(C)}$$

For bottom surface $\overrightarrow{dS} = -\rho\,d\rho\,d\varphi\,\vec{a}_z$

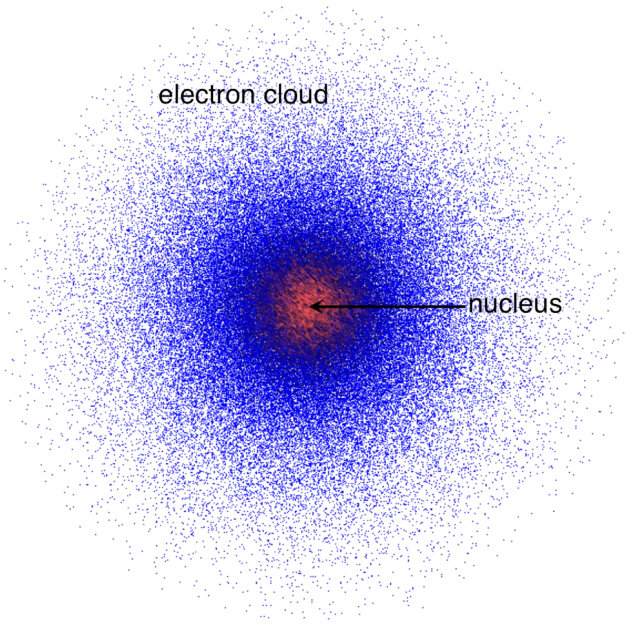
$$Flux = \iint_{Top}\vec{G}\cdot\overrightarrow{dS} = -10e^{-2(0)}\int_0^1\rho d\rho\int_0^{2\pi}d\varphi = -10\pi \quad \text{(C)}$$

For side surface $\overrightarrow{dS} = \rho\,d\varphi\,dz\,\vec{a}_\rho$

$$Flux = \iint_{side}\vec{G}\cdot\overrightarrow{dS} = 10e^{-2z}\rho^2\,d\varphi\,dz = 10(1)^2\int_0^{2\pi}d\varphi\int_0^1 e^{-2z}dz = 10\pi(1-e^{-2}) \quad \text{(C)}$$

Examples

(3) A cloud of electron with density of $\rho_v = -\frac{3\times10^{-8}}{r^4}\cos^2\varphi$ (C/m^3), exists between r=2 and 5 cm. Find the total charge.



$$\text{Volume Integral} \equiv \iiint_V (density)\, dv$$

Total charge: $Q = \iiint_V -\frac{3\times10^{-8}}{r^4}(\cos^2\varphi)\, r^2 \sin\theta\, dr\, d\theta\, d\varphi$

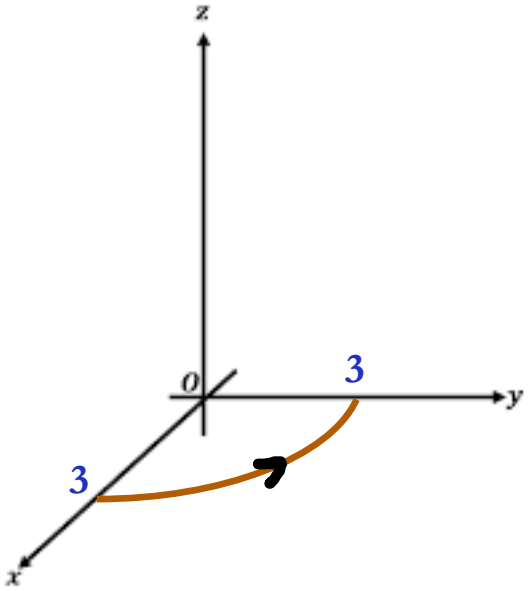
$$Q = -3\times10^{-8}\int_{0.02}^{0.05}\frac{r^2}{r^4}dr\int_0^{\pi}\sin\theta\, d\theta\int_0^{2\pi}(\cos^2\varphi)\, d\varphi$$

$$Q = -3\times10^{-8}\left[\left.\frac{1}{r}\right|_{0.02}^{0.05}(-\cos\theta)\Big|_0^{\pi}\left.\left(\frac{\varphi}{2}+\frac{\sin 2\varphi}{4}\right)\right|_0^{2\pi}\right]$$

$$Q = -1.8\pi\times10^{-6} \quad \text{(C)}$$

Examples

(4) Given $\vec{F} = xy\,\vec{a}_x - 2x\,\vec{a}_y$, find the line integral along the indicated path.



Transform to cylindrical coordinate system.

Perform line integral in cylindrical coordinate system (only φ component)

Answer: $-9\left(1+\frac{\pi}{2}\right)$

Vector Differentiation: Del Operator

The "del" operator ($\vec{\nabla}$) is the vector differentiation basic operation. It represents the rate of change with respect to space. It is defined in rectangular, cylindrical and spherical coordinates as follows.

Rectangular coordinate $$\vec{\nabla}= \frac{\partial}{\partial x}\,\vec{a}_x + \frac{\partial}{\partial y}\,\vec{a}_y + \frac{\partial}{\partial z}\vec{a}_z$$

Cylindrical coordinate $$\vec{\nabla}= \frac{\partial}{\partial \rho}\,\vec{a}_\rho + \frac{1}{\rho}\frac{\partial}{\partial \varphi}\,\vec{a}_\varphi + \frac{\partial}{\partial z}\vec{a}_z$$

Spherical coordinate $$\vec{\nabla}= \frac{\partial}{\partial r}\,\vec{a}_r + \frac{1}{r}\frac{\partial}{\partial \theta}\vec{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial}{\partial \varphi}\,\vec{a}_\varphi$$

Being a "vector" operator, it can be used with several possibilities.

Gradient	Divergence	Curl	Laplacian	Laplacian
$\vec{\nabla}A$	$\vec{\nabla}\cdot\vec{A}$	$\vec{\nabla}\times\vec{A}$	$\vec{\nabla}^2 A$	$\vec{\nabla}^2\vec{A}$

Vector Differentiation: Gradient

The gradient represents the three-dimensional "rate of change" of a given scalar function. So, it generates the rate of change vector.

Rectangular coordinate

$A(x, y, z)$ is a scalar function

$$\vec{\nabla} A = \frac{\partial A}{\partial x}\,\vec{a}_x + \frac{\partial A}{\partial y}\,\vec{a}_y + \frac{\partial A}{\partial z}\,\vec{a}_z$$

Cylindrical coordinate

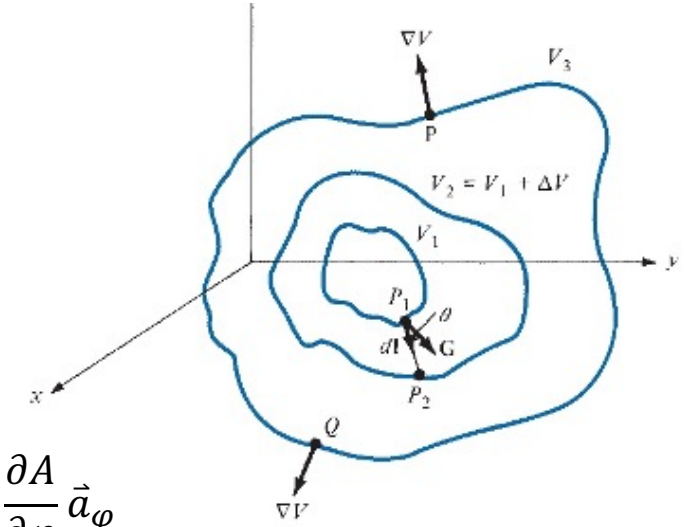
$A(\rho, \varphi, z)$ is a scalar function

$$\vec{\nabla} = \frac{\partial A}{\partial \rho}\,\vec{a}_\rho + \frac{1}{\rho}\frac{\partial A}{\partial \varphi}\,\vec{a}_\varphi + \frac{\partial A}{\partial z}\,\vec{a}_z$$

Spherical coordinate

$A(r, \theta, \varphi)$ is a scalar function

$$\vec{\nabla} = \frac{\partial A}{\partial r}\,\vec{a}_r + \frac{1}{r}\frac{\partial A}{\partial \theta}\,\vec{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial A}{\partial \varphi}\,\vec{a}_\varphi$$



Remarks:

- The gradient denotes the direction of greatest change of a scalar function.
- Its magnitude indicates how fast is the function changes in that direction.
- The gradient vector is always normal to equi-valued surfaces.

Example

The electrostatic field intensity is given by the gradient of the electric potential function. Find the electric field intensity at P(1, 1, 0). $\vec{E} = -\vec{\nabla}V$

Electric potential $V(x,y,z) = V_o e^{-x} \sin\frac{\pi y}{4}$ (V)

$$\vec{E} = -\vec{\nabla}V = -\left(\frac{\partial V}{\partial x}\,\vec{a}_x + \frac{\partial V}{\partial y}\,\vec{a}_y + \frac{\partial V}{\partial z}\vec{a}_z\right) = -\left(-V_o e^{-x}\sin\frac{\pi y}{4}\vec{a}_x + \frac{\pi}{4}V_o e^{-x}\cos\frac{\pi y}{4}\vec{a}_y\right)$$

$$\vec{E}_P = \frac{V_o}{e\sqrt{2}}\left(\vec{a}_x - \frac{\pi}{4}\vec{a}_y\right) \quad \text{(V/m)}$$

Electric potential $V(r,\theta,\varphi) = V_o r\cos\theta$ (V)

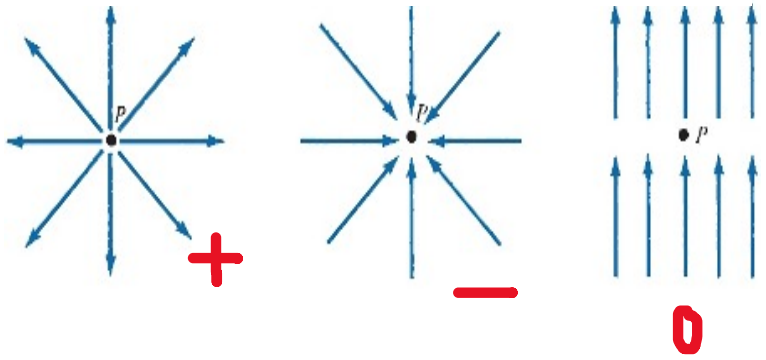
$$\vec{E} = -\vec{\nabla}V = -\left(\frac{\partial}{\partial r}\,\vec{a}_r + \frac{1}{r}\frac{\partial}{\partial\theta}\vec{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\vec{a}_\varphi\right) = -(V_o\cos\theta\,\vec{a}_r - V_o\sin\theta\,\vec{a}_\theta)$$

$$\vec{E}_P = \frac{V_o}{\sqrt{2}}\,\vec{a}_\theta \quad \text{(V/m)} \qquad \text{P(1, 1, 0)} \equiv \text{P}\left(\sqrt{2}, \frac{\pi}{2}, \frac{\pi}{4}\right)$$

Vector Differentiation: Divergence

The divergence operation measures how much a vector field is converging at (accumulating, sink) or diverging from (dispersing, source) a given point.

The result is a scalar quantity that is positive for source and negative for sink.



Rectangular coordinate

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$A(x, y, z)$ is a scalar function

Cylindrical coordinate

$$\vec{\nabla} \cdot \vec{A} = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho A_\rho\right) + \frac{1}{\rho}\frac{\partial A_\varphi}{\partial \varphi} + \frac{\partial A_z}{\partial z}$$

$A(\rho, \varphi, z)$ is a scalar function

Spherical coordinate

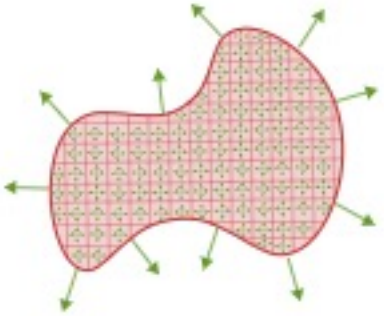
$$\vec{\nabla} \cdot \vec{A} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r \sin\theta}\frac{\partial}{\partial \theta}(A_\theta \sin\theta) + \frac{1}{r \sin\theta}\frac{\partial A_\varphi}{\partial \varphi}$$

$A(r, \theta, \varphi)$ is a scalar function

Divergence Theorem

$$\int_V \vec{\nabla} \cdot \vec{A}\, dv = \oint_S \vec{A} \cdot \overrightarrow{dS}$$

Sum of all divergence of points inside a volume is equal to the flux out of the volume



Examples

(1) Does the vector field $\vec{D} = 2xy\,\vec{a}_x - 7\,\vec{a}_y + (x^2 + z^2)\,\vec{a}_z$ behave as a source or a sink at P(0, 1, 3)?

$$\vec{\nabla}\cdot\vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = \frac{\partial 2xy}{\partial x} - \frac{\partial 7}{\partial y} + \frac{\partial (x^2+z^2)}{\partial z} = 2y + 2z$$

$$\left(\vec{\nabla}\cdot\vec{D}\right)\Big|_P = 8 \qquad \text{(Source)}$$

(2) The magnetic flux density due to a straight thin infinite wire carrying current I, at any distance ρ from the wire, is given by

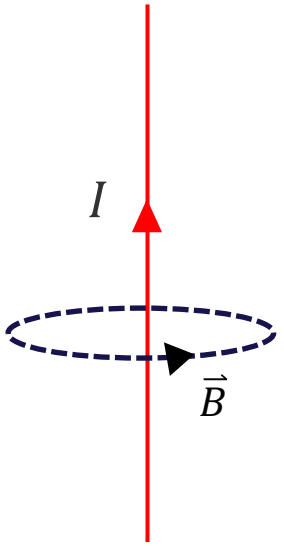
$$\vec{B} = \frac{\mu_o I}{2\pi\rho}\vec{a}_\varphi$$

Describe the field lines.

$$\vec{\nabla}\cdot\vec{B} = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho B_\rho) + \frac{1}{\rho}\frac{\partial B_\varphi}{\partial\varphi} + \frac{\partial B_z}{\partial z} = \frac{1}{\rho}\frac{\partial}{\partial\varphi}\left(\frac{\mu_o I}{2\pi\rho}\right) = 0$$

$$\vec{\nabla}\cdot\vec{B} = 0 \quad \Rightarrow \quad \int_V \vec{\nabla}\cdot\vec{A}\,dv = \oint_S \vec{A}\cdot\overrightarrow{dS} = 0$$

Divergence theorem



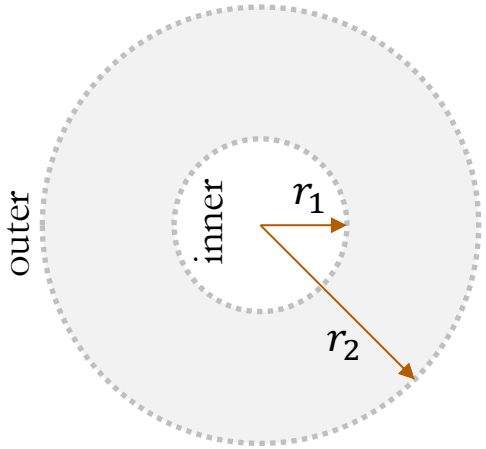
Magnetic field lines go around in circles

Examples

(3) Verify the divergence theorem of the vector $\vec{F} = kr\,\vec{a}_r$ over the given surface.

$$\vec{\nabla}\cdot\vec{F} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2F_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(F_\theta\sin\theta) + \frac{1}{r\sin\theta}\frac{\partial F_\varphi}{\partial\varphi}$$

$$\vec{\nabla}\cdot\vec{F} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2kr) = 3k$$



$$\int_V \vec{\nabla}\cdot\vec{F}\;dv = 3k\int_{r_1}^{r_2} r^2dr\int_0^{\pi}\sin\theta\;d\theta\int_0^{2\pi}d\varphi = 4\pi k(r_2^3 - r_1^3)$$

$$\oint_S \vec{F}\cdot\overrightarrow{dS} = \iint_{outer}\vec{F}\cdot\overrightarrow{dS} + \iint_{inner}\vec{F}\cdot\overrightarrow{dS}$$

$$\oint_S \vec{F}\cdot\overrightarrow{dS} = kr_2^3\int_0^{2\pi}d\varphi\int_0^{\pi}\sin\theta\,d\theta - kr_1^3\int_0^{2\pi}d\varphi\int_0^{\pi}\sin\theta\,d\theta = 4\pi k(r_2^3 - r_1^3)$$

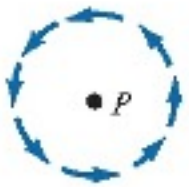
Vector Differentiation: Curl

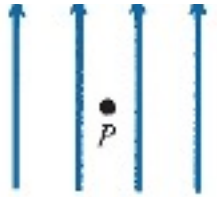
The curl operation measures how strong a vector field "circulates" around a point. The result is a vector that is normal to the circulation path.

Rectangular coordinate

$A(x, y, z)$ is a scalar function

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

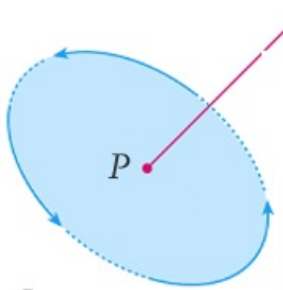




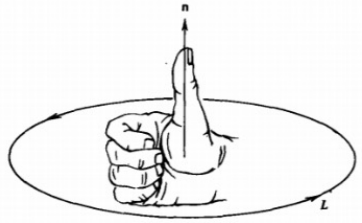
Cylindrical coordinate

$A(\rho, \varphi, z)$ is a scalar function

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho\mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

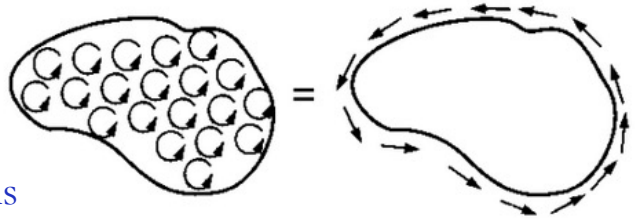


$$\nabla \times \mathbf{A} = \left[\frac{1}{\rho}\frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z}\right]\mathbf{a}_\rho + \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho}\right]\mathbf{a}_\phi + \frac{1}{\rho}\left[\frac{\partial}{\partial \rho}(\rho A_\phi) - \frac{\partial A_\rho}{\partial \phi}\right]\mathbf{a}_z$$



Stokes Theorem

$$\int_S (\vec{\nabla}\times\vec{A}) \cdot \overrightarrow{dS} = \oint_L \vec{A} \cdot \overrightarrow{dl}$$



Sum of all curl magnitudes of points over a surface is equal to the line integral around the surface boundary.

Q & A

Send me an email if you have any questions.